Autumn Pereira

Education

Cornell University — Ithaca, NY, USA
Doctor of Philosophy- School of Integrative Plant Sciences — *Aug. 2023 – Present*

McGill University — Montreal, QC, CA
Bachelor of Science- Joint Major in Biology and Mathematics — *Aug. 2019 – May 2023*

Research Interests

My research integrates metagenomic and amplicon sequencing approaches with direct measurements of greenhouse gas fluxes and other ecosystem function responses across diverse environments. I use these paired datasets to understand how microbial community composition and diversity regulate biogeochemical cycling.

Teaching

Invited Speaker – GEOG 495: Field Studies- Physical Geography — *Summer 2026*
McGill University

Undergraduate Course Assistant – MATH 242: Analysis I — *Fall 2021*
McGill University

Undergraduate Course Assistant – CHEM 120: General Chemistry 2 — *Winter 2021*
McGill University

Mentorship

Undergraduate Mentees — *2023 – Present*
Cornell University

- K-Shima Noble (2026-Present)
- Chloe Padelara (2025-Present)
- Letiunasema Nuusila (2023-Present)
- Jada Ihejirika (2023-Present)

Publications

Pre-Prints

Pereira, A., Martínez-Jerónimo, F., Fewer, D. P., Simon, D. F., Hernández-Zamora, M., Martínez-Jerónimo, L., Antuna-González, P., Munoz, G., Sauvé, S., Shapiro, B. J., & Tromas, N. "Linking Cyanobacterial Genomes to Toxin Dynamics Through Genome-Resolved Metagenomics". (2026) In: *bioRxiv* (p. 2026.02.24.707707) https://doi.org/10.64898/2026.02.24.707707

Peer-Reviewed Articles

Liao, C., Kao-Kniffin, J., Reid, M., Zhang, W., Tshabalala, P., Andrus, E., Butler-Jones, Z., LaPoint, J., Zhou, S., Rafols, R., Shin, S., Ichihara, M., **Pereira, A.**, Butler, N., Tu, Y., & McCouch, S. R. (2025). "Rice farming for climate change adaptation in the Northeastern United States. Proceedings of the National Academy of Sciences". (2025) In: *Proceedings of the National Academy of Sciences*, 122(49), e2402181122. https://doi.org/10.1073/pnas.2402181122

Tromas, N., Simon, D. F., Fortin, N., Hernández-Zamora, M., **Pereira, A.**, Mazza, A., Pacheco, S. M., Levesque, M.-J., Martínez-Jerónimo, L., Antuna-González, P., Munoz, G., Shapiro, B. J., Sauvé, S., & Martínez-Jerónimo, F. "Metagenomic insights into cyanotoxin dynamics in a Mexican subtropical lake". (2025) In: *Chemosphere*, 376, 144285. https://doi.org/10.1016/j.chemosphere.2025.144285

Presentations

Poster Presentations

Pereira, A., Kao-Kniffin, J. ”Increased GHG Emissions from organically managed rice and consequences for climate resilience agriculture in NYS”. (2025) Presented at: Cornell College of Agriculture and Life Sciences Innovation Day.

Pereira, A., Tromas, N. Shapiro, B.J., Sauve, S., Simon, D., Martínez-Jerónimo, F., Fortin, N., & Greer, C. ”An Investigation into the Strain-Level Dynamics of Microcystis and Implications for Microcystin Concentration”. (2023) Presented at: Interdisciplinary Freshwater Harmful Algal Blooms Workshop.

Awards

NSERC Postgraduate Scholarship – Doctoral *2024- 2027*
Merit-based fellowship covering 3 years of graduate stipend.

Cornell University Fellowship *2023- 2024*
Merit-based fellowship covering 1 year of graduate stipend.

Dean’s Multidisciplinary Undergraduate Research List *2023*
Distinction upon graduation from McGill for engagement in multidisciplinary undergraduate research.

Grants

Towards Sustainability Foundation Grants *2025*
Amount Awarded: $10,960.40 ||Role: Principal Investigator

Atkinson Center for Sustainability Moonshot Seed Grant *2024*
Amount Awarded: $46,356.00 ||Role: Primary Author

Service

Expanding Youth Horizons Science Outreach Event Volunteer *2026*
Cornell University

Cornell University Rice Field Day *2024*
Cornell University

Technical Skills

Bioinformatics: Amplicon sequence processing (DADA2), metagenomic assembly and annotation, strain-level profiling (inStrain, StrainGE), phylogenetic analysis

Computing: R, bash, Python, JavaScript (Google Earth Engine), Git, Slurm (HPC Cluster), LaTeX

Statistics: Multivariate statistics (PERMANOVA, PCoA, RDA), differential abundance analysis, co-occurrence network analysis, generalized linear modelling, time series analysis, non-parametric modelling

Field and Laboratory Methods: Static flux chamber measurements, gas chromatography, isotope ratio analysis, vegatation surveys, DNA extraction, PCR, amplicon library preparation